

TOTAL RECALL

Memory upgrades yield sweet performance increments

The RAM in your PC can be visualised as a highway between the most critical system components—the processor, video subsystem, and memory controller (if applicable).

How much memory is enough memory? The answer keeps changing every few months: while XP works well with 1 GB, the upcoming Windows Vista promises to reward PCs with 2 GB and above. Buying memory today requires careful thought, not only about performance increments but also future-proofing.

MYTHS AND REALITIES

☑ Better a new PC than new memory: True to a small extent, but look at the cost involved. The fact is, a memory upgrade can spruce up an older PC with a performance hike of up to 35 per cent!

☑ All memory with heat spreaders is good: While it's true that fast, low latency memory sticks come with heat spreaders, all value memory doesn't require such cooling solutions, and may heat up faster with such heat spreaders

due to lack of ventilation. What happens with ultra-low-latency memory is that the heat generated by the chips is transported away by the heatsink solution. In the case of value memory (which doesn't heat up as much), the heatsink prevents air circulation, which can actually cause the memory to start overheating.

☑ Memory is memory: Totally untrue! Never buy cheap or unbranded memory. Some companies charge you more partially for the brand, but more importantly because of the use of better quality and reliable memory chips on the RAM sticks. Most BSODs (Blue Screens of Death) and other freeze-up problems are caused by errors in memory banks—an indicator of cheap memory.

QUESTIONS TO ASK

What applications will I use?

Applications dictate memory requirements. Games require lots of memory, as does image editing, video editing, and 3D design and rendering. 2 GB is recommended in these cases. Audio encoding,

DVD playback, and heavy Internet browsing should get along fine with 1 GB. In all cases, consider 512 MB the minimum.

DDR2 or DDR?

DDR2 is pretty much *de facto* these days. If you're assembling a new PC, look only at DDR2. Due to high prices and scarcity, we don't recommend more DDR memory as a PC upgrade. It's better to save up and get a new motherboard, processor, and DDR2 memory later.

Single 1 GB stick or two 512 MB ones?

All motherboards today come with dual-channel support, which effectively



What To Look For

Lifetime Warranty: Most memory today comes with a lifetime warranty—product lifetime, not yours! In most cases, the warranty will be three years. Make sure you buy RAM with at least 3-year warranty.

Rated speed: Has the biggest impact on performance. Opt for at least 533 MHz for DDR2 memory in general. For one of the latest processors, you might need slightly faster memory to avoid it becoming a bottleneck. In such cases we recommend 667 MHz DDR2. If you absolutely have to upgrade your older DDR memory based PC, opt for DDR 400. All Intel LGA 775 socket motherboards will support DDR2 (with very few exceptions), while AMD's 939 processors support DDR memory. AMD's AM2 processors only support DDR2.

Latency Timings: Affects performance by as much as 15 per cent. The lower the timing values (typically, timing i.e. latency figures will have four numbers with dashes in between like 4-4-4-12), the better the memory will perform at the clock speed specified. Very-low-latency memory is expensive, while value RAM has higher timings. Typically, enthusiast (read low-latency) memory will have the timings boldly mentioned on the packaging. For DDR2 667 MHz, a latency of 4-4-4-12 or lower should be good enough. For DDR2 533, look for 3-4-4-8. DDR 400 MHz memory should have a timing of 3-3-3-8, if not lower. Keep in mind that latency figures vary proportionately with memory frequency.

doubles memory bandwidth. Enabling dual-channel mode requires two memory modules with identical clock speeds and densities, and ideally, identical latency timings and memory chips. In such a situation, two 512 MB sticks will perform much better than a single 1 GB stick.

Future Trends

The memory market has been in a state of flux: DDR is costly because it's en route to a phase-out. DDR2 has experienced explosive demand thanks to the acceptance of platforms supporting it. We should see prices stabilising soon. 2007 promises the debut of DDR3 memory. There are rumours about DDR3 replacing DDR2 as main memory, and the new DDR4 and XDR RAM being used as graphics memory soon. ■